image should be divided into three equal horizontal parts: forehead till eyes, nose, and lips till the chin.

Body language. This is a very important aspect, and there are many things involved in body language. But the main things to consider are walking pace, hand and leg movements, and postures.

For a more accurate analysis, it is important to find the correlation between facial expressions and body language for decision-making.

How to begin

Video clip is captured in the CCTV camera that is near the lift. It is called the unstructured form of data. This must be read by the machine in a different manner. Here comes the magic of deep learning, which is the most suitable technique in decoding any form of complex unstructured data.

How it is decoded

The core of AI is where we extensively use statistical and complex mathematical formulae and concepts like calculus, integration, derivatives, complex numbers and algebra fundamentals to solve this, and get results.

The video clip is broken into continuous shots of images and then every image is read by applying the above mathematical rules. All these rules are defined and calculated in API functions created by Python.

Every image can be divided into number of pixels. In a normal scenario, a coloured image is divided into 32×32 pixels. But since we have to divide the image in three parts, let us set the pixels as a round figure to make calculations simple for this scenario. Let us make the image in 30×30 pixels, so we get total 900 pixels for the image.

Now, we have the following ways an image is divided:

1. Every image is divided into three equal horizontal parts based on facial expressions, as explained earlier. This way we have 300 pixels for each part of the image.
2. Every image is a coloured image; thus, it is divided into three red, green and blue (RGB) images. Here, every colour will have 900 pixels. Thus, the total for three primary colours will be 2700 pixels.

This way we break down every image into 900 instances of weight, which forms the first layer in the deep learning mechanism.

Considering different formulae and calculations, we will keep adding different layers, and the last layer (output layer) will contain the total number of emotions captured and output using the deep learning mechanism—in this case, output will be seven emotions.

Let us divide the image in three parts and analyse each part separately. All three parts will enter the first layer together but will be weighted separately. So, we will have the following in the first layer.

First layer preceptor: This is divided into a square and very small pixels; in our case, 30×10 pixels (one third of the image is taken). The higher the number of pixels, the more accurate the results. Every pixel, depending on darkness, will give numbers ranging from 0.0 to 1.0. These are called weights of every pixel.

Similarly, the other two images will also be weighed.

Second layer preceptor: All pixel numbers will be captured and given to the second layer.

This will capture every single pixel based on the higher or lower number, and finally the figure will be read. Here we can capture lines formed near the eyes. Based on the lines and shape of the eyes captured we can decide the emotion.

It is the same for the other two images.

Now we get three different weights of eyes, nose and lips, which will help us come to some conclusion.

Drop-out layer: When we have created five to six layers, and we come across some layer that shows too much learning or very little learning, both of which are not so useful in machine learning and deep learning, then that layer must be dropped out of the structure. But this must be done after proper analysis and understanding, and it depends completely on the data scientist and the situation to decide which layer must be dropped.

Back propagation analysis

This figure can be compared with back propagation of different emojis that we have. This analysis is done when we get some wrong result at the training stage. If we show a happy face and its giving output as sad face, then we have to do back propagation and find the source of the issue, and then solve it.

In our case, there will be an output of seven different emojis, and in the training data, we give all seven images for the machine to learn. This step is sort of reverse engineering, where we go back from the final layer to the previous layer to check if there are any mistakes.

Once that is resolved, we can move ahead with test data, which will be the actual image that will be used for machine learning to figure out the emotion. If we do this correctly, with hundred per cent results, then in test data we should get around 95 per cent correct result.

If this is successful, we have a ready machine learning mechanism for facial expressions and face reading software. This will help us analyse the face expressions and decide if these are happy or sad ones. END

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